A lesson-plan-driven framework for generating learning materials: a Nigerian mathematics case study

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Abstract

Preparing differentiated learning material is time-intensive for teachers, and large language models offer a way to reduce that cost. Existing systems, however, take curriculum standards or a seed task as the generation specification, so a generator is left to infer the lesson a class actually received. We propose a lesson-plan-driven framework in which the artefacts a teacher already produces are the specification: it composes curriculum objectives, the teacher's lesson plan, subject knowledge retrieved under scope constraints, and learner state from marked class work, and admits nothing to a learner without teacher review. We instantiate it on Nigerian junior-secondary mathematics and compare cumulative configurations over 72 generated packs. Four practising teachers judged the material usable — 95 of the fully configured condition's 101 instructional-quality ratings at 4 or above — and rated it higher on alignment with the planned lesson than objectives-only generation, favourable ratings rising from 81.0% to 92.2% and doubly rated sections with both raters favourable from 33 of 49 to 44 of 51, though six lessons cannot take that below $p = .06$ and no component is separately established. The learner rung adapted exactly the objectives it held evidence for and left the others byte-identical, with raters identifying targeted error support in particular packs but no consistent preference over the class pack. A blinded judge, reported separately, orders the configurations the same way. The contribution is an implemented framework that makes teacher-authored instructional knowledge explicit in educational generation, with a locally situated evaluation of both its rungs.

1 Introduction

Producing the material a lesson requires — explanation, worked examples, practice and assessment — is a substantial, recurring part of a teacher's workload [Ajayi and Ajayi, 2015, Yao et al., 2025], and it is typically authored once for a whole class. Differentiated instruction asks for more: that content, process and product vary with learners' readiness, interest and pace [Tomlinson, 1999], an expectation personalised-learning research has since made explicit [Bernacki et al., 2021]. That expectation is additive: it does not replace existing preparation but multiplies it by class size, and must be met inside curricula prescribing what is covered and when [Nikolaidis et al., 2024]; teachers accordingly want AI to complement rather than replace their judgement [Seo et al., 2024].

Large language models can supply part of that capacity, generating explanations [Leinonen et al., 2023], worked examples [Jury et al., 2024] and controllable practice questions [Li and Zhang, 2024], with Sharma et al. [2025] surveying the effect on personalised learning. What is missing is an account running from planning to delivery. Existing systems take curriculum standards or a seed task as the generation specification [Liu et al., 2025, Gonnermann-Müller et al., 2025, Sesay et al., 2026], so a generator must infer how the topic was approached, in what order, and against which anticipated

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

errors; the material reflects the curriculum rather than the lesson a class received. Nor is it settled how learner evidence should enter such a pipeline [Cho et al., 2024].

Teachers already produce the missing specification. Planning is routine, documented practice [Yinger, 1980, Musingafi et al., 2015]: a lesson plan precedes each lesson, a lesson note derives from it, and marked class work accumulates as teaching proceeds. We ask whether these artefacts can drive generation of downstream material, and whether the marked work can then personalise it, without additional authoring. We address this through design science research [Hevner et al., 2004, Peffers et al., 2007], in which the contribution is a designed artefact whose objectives are inferred from the problem. The artefact is a lesson-plan-driven framework, adapting the component decomposition of classical intelligent tutoring systems [Wenger, 1987, Nwana, 1990]: it separates curriculum objectives, a lesson-plan specification, scope-constrained subject knowledge, and learner state from marked class work, and admits nothing to a learner without teacher review.

Two design objectives follow, which Hevner et al. [2004] require be demonstrated rather than asserted, so we restate them as testable claims. **H1**: material generated under a lesson-plan specification aligns better with the taught lesson and its curriculum objectives than material generated from curriculum objectives alone. **H2**: material generated under a learner state derived from marked class work addresses an individual pupil's learning better than generic class-level material. We evaluate both. Our contributions are:

- **A formal lesson-plan-driven generation framework.** We represent teacher-authored lesson plans as explicit instructional specifications and integrate them with curriculum objectives, subject knowledge, target-material requirements and objective-level learner state in an ITS-inspired generation-and-validation framework under teacher review.
- **A situated evaluation of usefulness and teacher alignment.** We instantiate the framework in Nigerian JSS1 mathematics and compare cumulative configurations across 72 generated packs, giving evidence of educator-assessed usefulness and higher observed teacher-planned lesson alignment in the fuller configurations than in objectives-only generation.
- **A demonstration and evaluation of selective personalisation.** We evaluate learner-specific adaptation across 26 learner-lesson cases, verify preservation of sections outside the recorded adaptation scope, and document educator-identified examples of targeted error support, distinguished from the stronger claim of a consistent advantage over class-level material.

2 Related work

Two lines of work are relevant. The first generates learner-facing material from a curriculum specification. Liu et al. [2025] propose COGENT, producing grade-appropriate passages from NGSS concepts; Gonnermann-Müller et al. [2025] present FACET, generating a differentiated worksheet from a seed problem with a learner simulated from two binary traits; Sesay et al. [2026] describe LectūraAgents, which assembles lecture packages from a syllabus it writes itself. LearnLens couples error-aware assessment with curriculum-grounded generation under educator oversight [Zhao et al., 2025] and GenMentor schedules a learning path from a learner's goal [Wang et al., 2025]; Holstein et al. [2020] frame the division of adaptive work between teacher and system.

The second line generates *lesson plans*: Ranawat et al. [2021] sequence web-annotated plans, Moundridou et al. [2024] implement inquiry-based ones, Olaniyan et al. [2025] align generated plans to Bloom's taxonomy, Zhang et al. [2025] and Zheng et al. [2025] propose multi-agent and knowledge-grounded plan generation, and Celik et al. [2026] study pre-service teachers co-constructing plans with a model. Closest to our setting, Varuvel Dennison et al. [2026] report Shiksha Copilot, with which teachers in low-resource schools curate and customise generated plans. Our position is the converse: these systems treat the lesson plan as an *output*, whereas we treat an existing teacher-authored plan as the *input* constraining generation of other material. We are aware of no prior system that takes the plan in that role.

Two further gaps are methodological. These systems introduce several capabilities together and evaluate the assembled pipeline against a baseline, so the field has end-to-end results without evidence about which components produce them; we are aware of no published component-additive ablation of an educational LLM generation system. And the three closest comparators evaluate with expert human raters, yet none reports an inter-rater reliability statistic or validates its judge against its

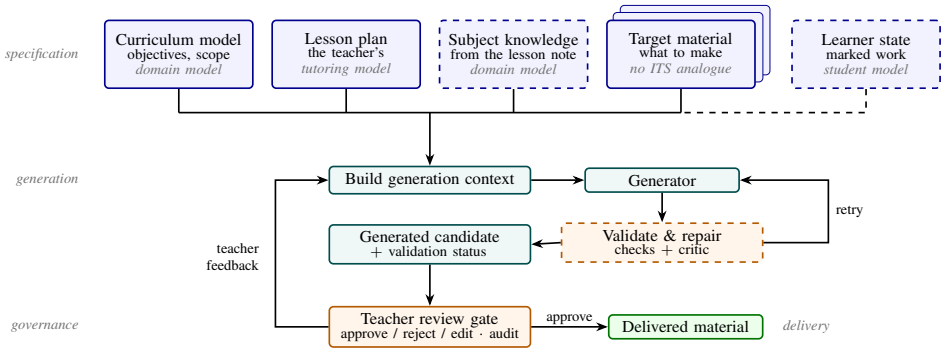


Figure 1: The framework. The tutoring components it adapts are named in italic, and dashed boxes are optional: an instance may omit retrieval, validation or the learner state. The target material is drawn as a stack because substituting it changes what is produced — a lesson note, a worksheet, an assessment — without changing the components that supply it. The target material and the review gate have no counterpart in that decomposition — they are what generating content, rather than selecting it, requires. The gate is a decision point: an approved pack is delivered, and teacher feedback returns the material to generation.

humans, and LectūraAgents assigns one rater per item, so reliability there is uncomputable. We measure ours in §6.2.

3 The framework

Teachers already record what they intend to teach and what their pupils did with it. The framework takes those records as the specification a generator works from: the plan says what the lesson was, the note carries its subject matter, marked class work the evidence about each pupil. The generator, the source retrieval reads and the kind of material produced are left open.

Five design principles fix what any implementation must satisfy. The lesson plan is the binding specification, constraining content to its scope and objectives (**DP1**), since curriculum standards alone leave the generator to infer the lesson. Teacher approval precedes learner access, as a non-optional gate with an audit record (**DP2**), and personalisation is constrained so that difficulty may vary but topic scope may not (**DP4**): teachers ask that AI complement their judgement [Seo et al., 2024] inside a curriculum prescribing what is covered and when [Nikolaidis et al., 2024]. Every item carries an objective tag and a provenance link (**DP3**), so material can be checked against the lesson it serves. And generation happens ahead of time, separating model-dependent preparation from learner use (**DP5**), for a school that cannot depend on a model at the point of use; this last we do not evaluate. Only DP1 and DP2 are structural: an instantiation without the lesson plan or the review gate is not an instance of this framework, while retrieval, validation and the learner state may be present or absent.

Four layers realise those principles (Figure 1). The **specification** layer holds the objects generation is conditioned on. In the classical decomposition [Wenger, 1987, Nwana, 1990] a domain model represents the subject matter and what counts as correct performance; here retrieval supplies the former and the curriculum the latter. Learner state is the student model, and the lesson plan stands in for a tutoring model the system would otherwise infer. The layer also imposes four curriculum constraints on the output: *scope* (only units in the current period), *coverage* (every objective addressed), *resource* (only mapped sources) and *sequence* (prerequisites from earlier periods). The **generation** layer projects those objects into a request, drafts and validates it; the **governance** layer decides what reaches a learner; and **delivery** sits outside the generation path.

3.1 The specification layer

Systems that generate educational content typically condition on curriculum objectives. Let a curriculum unit carry an ordered set of objectives $O = (o_1, \ldots, o_n)$, each a sentence of the form *carry out order of operation in algebra*. An objective specifies what a learner should be able to do at the end of a unit, but not how instruction reached that outcome.

A lesson plan specifies the latter. Written before the lesson and structured by the national format, it records a teacher's decisions about one session. Yinger [1980] describes planning as a staged process in which a teacher formulates and resolves an instructional problem before delivery, and Biggs [1996] argues that alignment between objectives, activities and assessment is what makes such planning instructionally effective. The plan is where both are recorded. We represent it as

$$P \;=\; \langle\, O_P,\; S,\; A,\; E,\; K,\; M \,\rangle, \tag{1}$$

where O_P are the lesson objectives the teacher derived from O, S an ordered sequence of instructional steps, A the assessment she planned, E the entry behaviour she assumes, K the misconceptions she anticipates and M her materials. Each step is $s = \langle \rho, \kappa, o, T \rangle$: a structural role $\rho \in \{\text{INTRO}, \text{CORE}, \text{EVAL}\}$, the kind of knowledge $\kappa \in \{\text{CONCEPT}, \text{PROCEDURE}, \text{REPRESENTATION}\}$ a core step teaches, the objective $o \in O_P$ it serves, and the activities T it sets. P therefore differs from O in kind rather than in detail: a sequenced, typed record of the instruction planned to reach the outcomes O names.

Two further objects complete the layer. The target material is a first-class input, so the framework is not tied to one output type; adaptation additionally requires learner evidence, for which we use the marked class work a school already holds. For pupil ℓ and objective o,

$$\Sigma = \langle\, \alpha,\; \beta,\; R \,\rangle, \qquad \Lambda(\ell, o) = \langle\, b,\; m,\; c,\; d,\; \mu,\; V \,\rangle, \quad b \in \{\text{REMEDIATE}, \text{REINFORCE}, \text{EXTEND}\}. \tag{2}$$

Σ names the material kind α, its audience β and the requirements R it must satisfy; substituting it changes what is produced — a lesson note, a revision worksheet, an assessment — without changing the components that supply it. In Λ, b is a mastery band, m the evidence supporting it (a count over items, not a bare fraction), c and d the pupil's confidence and perceived difficulty, μ any diagnosed misconception and V the marked answers. Λ is indexed by objective, not by pupil: a learner may need extension on one objective and remediation on the next.

3.2 The generation layer

A plan is not supplied to the model in full. A policy θ projects it onto the input for a single output section,

$$\pi_\theta : (P, o) \;\longmapsto\; U, \tag{3}$$

declaring field by field what travels. In our instantiation θ codes the teacher's narration as instructional events and drops the raw prose, carries lesson-level materials on the lesson context rather than per step, and withholds step titles because most restate the objective. Generation is then $G(\Sigma, U) \mapsto Y$, producing a section Y against the objective it addresses, and $\mathcal{V}$ validates the draft against the constraint schema.

Two operators complete the layer. Scoped retrieval $\varrho(o)$ adds evidence for the objective, and a validation-and-repair operator $\mathcal{V}$ checks the draft against the constraint schema and repairs what fails. Because θ is declared rather than implicit, an instance's inputs can be varied one at a time — which is what makes the design evaluable component by component.

3.3 The governance layer

Nothing reaches a learner without teacher approval, and the gate is a decision rather than a formality: a pack is approved, rejected, or returned to generation with the teacher's feedback, and the decision is recorded with its reason (**DP2**). A returned pack re-enters as a revision turn constrained to the same objective set, so review can change what a section says but cannot quietly drop an objective the plan requires. Delivery is separate again: packs are compiled ahead of use (**DP5**), so the model is needed when a teacher prepares, not when a pupil reads.

3.4 Binding and cost

The framework's terms bind to a local curriculum vocabulary: in the Nigerian NERDC curriculum a unit is a subtopic and objectives are performance objectives, the basis of the claimed reusability and an argument from structure, since we evaluate one binding here. Constraints are *supplied* and the output *checked* against them — a mechanism, not a guarantee, since schema validity establishes

neither mathematical correctness nor faithful sequencing — and teacher approval is a governance component, distinct from $\mathcal{V}$.

Manual differentiation scales with teacher time per pupil; here a second pupil's pack costs one further generation call over the same specification plus $\Lambda(\ell, o)$, from work already marked, with no new authoring and no live model at the point of use. Teacher effort shifts from authoring to review — a property of the design, not a measured saving.

4 Instantiation

We demonstrate the framework on Nigerian junior secondary mathematics (JSS1, pupils aged roughly 11–13) in a class of forty taught by one teacher, a setting with limited capacity for manual differentiation. Musingafi et al. [2015] describe how curriculum expectations pass through syllabi and schemes of work into the lesson plan, where a teacher fixes objectives, sequence and assessment for one session; in Nigerian schools Ajayi and Ajayi [2015] report uneven competence in preparing such lessons and Ayo-Vaughan [2022] variation in whether the prescribed formats are used, so a framework consuming plans must accept what teachers actually write. The study uses the materials the school produces for every lesson, a lesson plan and a lesson note, which the framework consumes directly.

The framework produces the material named by Σ. Here Σ specifies a post-class revision pack, a self-study document a pupil works through at home, and holds the output requirements fixed across conditions: its kind α is "one section of a take-home revision pack", its audience β a JSS1 pupil revising alone at home, and its five requirements R fix the register (never about the lesson as an event), the title, the topic scope, the presence of answers, and what a section may refer to. Substituting a different Σ — an exit ticket, a worksheet — changes the material without changing the components that supply it.

Each unit U is keyed to one objective, so a pack has one section per objective. Within π_θ the narration in T is coded as Gagné instructional events [Gagné, 1985] by a deterministic rule set: one code per activity, weak evidence uncoded, duplicates collapsed. Five of the nine events occur here and the other four are never inferred, so the coding describes the lesson the teacher planned, not one the framework recommends. For lesson t2-w09 and $o =$ *carry out order of operation in algebra*, $\pi_\theta(P, o)$ carries the step's position in S, the events GUIDANCE, PERFORMANCE and FEEDBACK, the four items she set ($4n + 3n \times 10$, $32 + 8x \div 2$, $3 \times 5x + 4x \div 2$, $36y \div 6 + y + 2 \times 4y$) and her assessment; under C0 the unit is $U_0 = \langle o \rangle$ and carries none of it. The C1 pack practises on her items, the C0 pack on three it generates itself ($3x + 2 \times 4x$, $5(2x + 3)$, $4x + 3(2x - 1)$) without the division hers exercise. Reuse of her exercises is not the criterion: newly generated exercises may equally satisfy Σ, and fidelity concerns the instructional requirements rather than reuse.

5 Evaluation design

The class-level ablation was evaluated on two independent instruments applied to the same 72 packs: a blinded LLM judge and a panel of practising teachers. We report both, rather than the one that separates.

Corpus. Six lessons were drawn from two terms of the teacher's own record, covering 17 learning objectives, and each of the four class-level settings generated three independent draws per lesson, giving $6 \times 4 \times 3 = 72$ packs. The corpus additionally holds the marked class work of nine pupils across three lessons, sampled by stratified random selection over the teacher's ability categories, which were not themselves recorded, so we report no stratum result. The stratum determines whom we sample; Λ responds to objective-level evidence.

Conditions. Each condition is a setting of the projection policy θ over one architecture, adding a component at a time (Table 1). Because Σ, the generator and the output contract are fixed, a rung's effect is that component's effect.

Criteria. The human panel scores three constructs: whether the material addresses the stated objective (*curriculum-objective alignment*); whether it reflects the lesson as planned (*lesson-specification*

Table 1: The ablation. Inputs are cumulative. C0 supplies the objective alone — the framework with its specification removed, and so not an instance of it. Retrieval is capped at three evidence items per unit with source references required; $\mathcal{V}$ runs eight structural checks and an LLM critic with up to two repair retries.

	Unit supplied to G	Component added	Evaluated here
C0	$U_0 = \langle o \rangle$	Objective only	yes
C1	$\pi_\theta(P, o)$	The teacher's lesson plan	yes
C2	$\pi_\theta(P, o) \cup \varrho(o)$	Scoped retrieval over the lesson note	yes
C3	C2's unit, then $\mathcal{V}$	Validation and repair	yes
P1	C3's unit + $\Lambda(\ell, o)$	Learner state from marked work	yes

fidelity, the primary construct, which operationalises H1); and whether it is mathematically correct, clear and useful (*instructional quality*), each 1–5 against fixed descriptors. The judge scores four dimensions on the same scale, quality split into correctness and clarity; we report the primary construct alone as well as the composite.

Primary comparison. The comparison of primary interest is the lesson-plan-bearing family (C1–C3) against the objectives-only baseline (C0), with diagnostics C1−C0, C2−C1 and C3−C2 locating any effect. The endpoint frozen before data collection — internally recorded, not publicly registered — was C1 versus C0 on the human panel; the family contrast was adopted afterwards, once judge scores had been inspected, and is exploratory, as are the diagnostics, whose p-values are unadjusted.

The judge. The LLM judge (Claude Sonnet 4.6) is blinded to condition and is constrained in code to differ from the generator. It was run twice over all 72 packs in fully independent passes, so its repeat reliability could be measured before its scores were interpreted.

The human panel. Four practising mathematics teachers rated packs on the three criteria per objective, blind to condition and with delivery order counterbalanced. We report their scores as levels, as condition means and favourable rates, and as what the two raters of a section agreed on, rather than through a model that would have to assume how rater variance enters.

Implementation and aggregation. All 72 packs were generated by DeepSeek-V4-Flash at temperature 0 with caching disabled, using a DSPy 3.3.0 harness [Khattab et al., 2023]: our own composition module driving a ReAct-style tool loop [Yao et al., 2023], evidence in-prompt, rewrite-style repair. DSPy is a programming framework here; no prompt-optimisation procedure was run. The same model and harness generated every condition. Per lesson and condition, judge scores average over three packs and both passes; C1–C3 are weighted equally within a lesson, compared with C0, and the six lesson differences averaged equally.

6 Results

6.1 The components executed

Before asking whether the rungs differ in quality we verify from the generation log that they differ at all, with no rater involved. Retrieval attaches source references to 168 blocks at C2 and 157 at C3, around 58%, and none at C0 or C1 where it is disabled; the validator ran on all 18 C3 packs and no others, logging 76 findings, a mean of 4.22 per pack.

6.2 What the human panel could resolve

The panel returned 1,212 of 1,224 planned scores, with delivery position balanced across conditions by design (ANOVA, $F = 0.01$, $p = .998$). Read as levels rather than differences, the material is judged usable: on instructional quality 95 of C3's 101 ratings are at 4 or above and both raters were favourable on 44 of its 50 doubly rated sections. C0 is high there too, at 93 of 100, so the panel does not separate the conditions on quality.

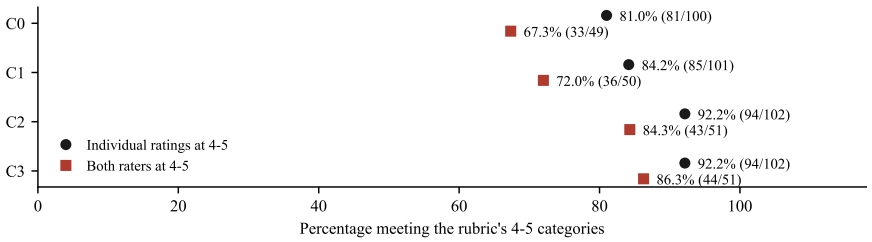


Figure 2: Teacher-planned lesson alignment, as individual favourable ratings and as sections both raters marked favourably. Denominators differ because 12 assigned ratings were not returned.

It separates them on the criterion H1 names (Figure 2). Across C0–C3 the lesson-context means run 4.26, 4.34, 4.53 and 4.58, and sections where either rater went to 2 or below fall 5, 2, 1, 0.

One difference per lesson puts C3 ahead of C0 in five of six lessons and level in the sixth, by $+0.283$; C2 by $+0.190$ (4/6), C1 by $+0.068$ (3/6), the equally weighted family by $+0.180$ (4/6). Six lessons put the exact signed-rank floor at $p = .0625$, where C3 lands: consistent across lessons, no single rung separately established.

Curriculum-objective alignment sits at 90–94% at 4–5 in every condition, which is why we read the focal criterion rather than the three-criterion composite — a decision taken after the protocol, which named the composite.

What limits the reading is the raters, not the material. On this criterion their means run 4.57, 4.47, 4.56 and 4.12 and their favourable rates 88.7%, 96.0%, 86.5% and 78.8%, and the two raters of a section differ by two points or more on 10 of C0's 49 sections, 9 of 50, 6 of 51 and 5 of 51. Disagreement falls as the configuration fills, but never to a level at which the panel could resolve a rung on its own.

It is a stable instrument: run twice in fully independent passes over the same 72 packs it moved a score by 0.076 on average and returned an identical score on 92.4% of 288 comparisons, with no justification string byte-identical between passes.

6.3 What the automated judge adds

On the judge the lesson-plan-bearing family scores above the objectives-only baseline in every one of the six lessons, exceeding C0 on the primary construct by $+0.750$ (Wilcoxon signed-rank, two-sided $p = .031$).

The diagnostics locate most of that difference at one rung: adding the lesson plan moves the primary construct $+0.694$ (5/6, $p = .0625$), retrieval a further $+0.111$ (3/6) and validation -0.056 (0/6). That is evidence the plan carries most of the family effect and no evidence of further contribution, not evidence the contribution is zero.

The difference is concentrated in low-score flags. Of the 18 objectives-only packs, 8 carry at least one judged score below 4; of the 54 generated with a lesson plan, none does (Fisher exact $p = 3.7 \times 10^{-6}$), an operationally defined flag rather than a confirmed defect.

They are construct-specific: 8 of 18 C0 packs on lesson-specification fidelity, against 1 on curriculum-objective alignment, 1 on conceptual correctness and none on clarity.

The validation component did not close. The validator closed its repair loop on only 6 of the 18 C3 packs; the other 12 exhausted both retries still carrying a finding, consistent with that rung showing no gain.

6.4 The learner-model rung

Part 2 put each pupil's personalised pack against the class base pack: 26 learner-lesson cases over nine pupils and three lessons, 35 objective panels rated by two teachers each on opposite orders, 199 of 210 cells answered.

Personalisation was selective, a property of the generation record rather than a judgement: compared section by section within an objective, P1 changed exactly the 35 objectives for which it held learner state and left the 34 it did not byte-identical.

The comparative judgement goes no further. Scoring each judgement $+1$ for P1, 0 for about the same and -1 for P0, and weighting pupils equally, P1 exceeds P0 by $+0.045$ (stratified cluster bootstrap over the nine pupils, $[-0.106, +0.199]$); the direction varies by criterion: P1 on 38 of 67 adaptive-calibration judgements, 25 of 67 for learner-evidence alignment against 24, and 20 of 65 for instructional integrity against 25. On adaptive calibration the two raters chose opposite materials on 11 of 32 doubly rated panels. Comments were optional; 15 cells carry one, and some identify targeted support — one records that the personalised section "repeatedly laid emphasis on the mistake and carefully broke the steps down". Instances, not a rate.

7 Discussion

Teachers judged the material usable and rated the fuller configurations higher on alignment with the lesson they had planned; the automated judge orders the configurations the same way. Neither instrument establishes a separate contribution for any one component — the increase concentrates where the plan enters, and retrieval adds no measured gain, though it attaches source references to roughly 58% of blocks, a traceability aggregate ratings were not designed to detect. On H2 the mechanism and the judgement come apart: personalisation was exactly as selective as the design intends, and teachers still did not prefer its output. Adoption also depends on teacher readiness, which Ayanwale et al. [2025] model for the Nigerian context.

8 Limitations

Five limitations bound these results. First, both instruments have known limits: the judge is validated for repeat stability but not against human judgement, and the panel's pooled agreement is weak, so neither result rests on a reference we can call reliable. Second, C0 is our own pipeline with its inputs removed, so the ablation establishes internal attribution, not external superiority. Third, the evidence does not travel far from the case that produced it — one subject, one grade, one school, six lessons, one generator — so the framework's reuse remains an argument from structure. Fourth, we evaluate material and not learning: there is no pre-test, post-test or retention measure, and H2's instrument asks which of two documents a teacher prefers, not whether either taught anything. Finally, we did not measure the workflow burden that motivates the framework: no instrument for preparation or verification effort was run.

9 Conclusion

We presented a lesson-plan-driven framework for generating curriculum-constrained learning material, in which the artefacts a teacher already produces serve as the generation specification. It separates the curriculum model, the lesson plan, the subject knowledge that grounds it and learner state from marked class work, and combines them through a compile-then-review process in which nothing reaches a learner without teacher approval.

Two properties follow from that separation. Because the target material Σ is a first-class input, the same components produce a lesson note, a revision worksheet or an assessment without alteration. And because the plan reaches the generator through a declared projection policy θ, a configuration is a setting of that policy rather than a distinct pipeline, which makes the design evaluable component by component.

The Nigerian JSS1 instantiation shows the framework operating on artefacts a school already produces: material teachers judged usable, ahead of objectives-only generation on the criterion H1 names on both instruments without reaching conventional significance at six lessons, and a learner rung that personalised exactly the objectives it held evidence for, without teachers preferring the result. A stronger human instrument and a wider binding are the next steps, against the proposition that educational generation can be organised around instructional knowledge teachers already produce.

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